

Mathematical Analysis II

Week 4 Homework (March 24)

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9. 在下列各题中, 求 $\frac{\partial^2 z}{\partial x^2}$, $\frac{\partial^2 z}{\partial x \partial y}$, $\frac{\partial^2 z}{\partial y^2}$

(2) $z = \arctan \frac{x+y}{1-xy}$;

Solution. 注意到 $z = \arctan \frac{x+y}{1-xy} = \arctan x + \arctan y$, 因此

$$\frac{\partial z}{\partial x} = \frac{1}{1+x^2}, \quad \frac{\partial z}{\partial y} = \frac{1}{1+y^2}.$$

从而

- $\frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{1}{1+x^2} \right) = -\frac{2x}{(1+x^2)^2}$
- $\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{1}{1+y^2} \right) = 0$
- $\frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{1}{1+y^2} \right) = -\frac{2y}{(1+y^2)^2}$

(4) $z = \sin^2(ax + by)$;

Solution. 将 z 化简为

$$z = \frac{1 - \cos(2ax + 2by)}{2},$$

因此

$$\frac{\partial z}{\partial x} = a \sin(2ax + 2by), \quad \frac{\partial z}{\partial y} = b \sin(2ax + 2by).$$

从而

- $\frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} (a \sin(2ax + 2by)) = 2a^2 \cos(2ax + 2by)$
- $\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x} (b \sin(2ax + 2by)) = 2ab \cos(2ax + 2by)$

- $\frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y} (b \sin(2ax + 2by)) = 2b^2 \cos(2ax + 2by)$

(6) $z = \arcsin(xy)$.

Solution. 一阶偏导数为

$$\frac{\partial z}{\partial x} = \frac{y}{\sqrt{1-x^2y^2}}, \quad \frac{\partial z}{\partial y} = \frac{x}{\sqrt{1-x^2y^2}}.$$

因此

- $\frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{y}{\sqrt{1-x^2y^2}} \right) = y \cdot \frac{xy^2}{(1-x^2y^2)^{\frac{3}{2}}} = \frac{xy^3}{(1-x^2y^2)^{\frac{3}{2}}}$

- $\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{x}{\sqrt{1-x^2y^2}} \right) = \frac{1}{(1-x^2y^2)^{\frac{3}{2}}}$

- $\frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{x}{\sqrt{1-x^2y^2}} \right) = x \cdot \frac{x^2y}{(1-x^2y^2)^{\frac{3}{2}}} = \frac{x^3y}{(1-x^2y^2)^{\frac{3}{2}}}$

10. 设 $u = e^{xyz}$, 求 $\frac{\partial^3 u}{\partial x \partial y \partial z}, \frac{\partial^3 u}{\partial x \partial y^2}$.

Solution. 对于 $\frac{\partial^3 u}{\partial x \partial y \partial z}$, 有

$$\begin{aligned} \frac{\partial^3 u}{\partial x \partial y \partial z} &= \frac{\partial^2}{\partial x \partial y} (xye^{xyz}) \\ &= \frac{\partial}{\partial x} (xe^{xyz} + x^2ye^{xyz}) \\ &= e^{xyz} + xye^{xyz} + 2xyze^{xyz} + x^2y^2ze^{xyz} \\ &= (1 + 3xyz + x^2y^2z^2)e^{xyz}. \end{aligned}$$

对于 $\frac{\partial^3 u}{\partial x \partial y^2}$, 有

$$\begin{aligned} \frac{\partial^3 u}{\partial x \partial y^2} &= \frac{\partial^2}{\partial x \partial y} (xze^{xyz}) \\ &= \frac{\partial}{\partial x} (x^2z^2e^{xyz}) \\ &= 2xz^2e^{xyz} + x^2yz^3e^{xyz}. \end{aligned}$$

11. 设 $r = \sqrt{x^2 + y^2 + z^2}$, 证明当 $r \neq 0$ 时有

(1) $\frac{\partial^2 r}{\partial x^2} + \frac{\partial^2 r}{\partial y^2} + \frac{\partial^2 r}{\partial z^2} = \frac{2}{r};$

Proof. 一阶偏导数为

$$\frac{\partial r}{\partial x} = \frac{x}{r}, \quad \frac{\partial r}{\partial y} = \frac{y}{r}, \quad \frac{\partial r}{\partial z} = \frac{z}{r}.$$

因此

$$\frac{\partial^2 r}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{x}{r} \right) = \frac{r - x \cdot \frac{\partial r}{\partial x}}{r^2} = \frac{r - x \cdot \frac{x}{r}}{r^2} = \frac{r^2 - x^2}{r^3}.$$

同理可得

$$\frac{\partial^2 r}{\partial y^2} = \frac{r^2 - y^2}{r^3}, \quad \frac{\partial^2 r}{\partial z^2} = \frac{r^2 - z^2}{r^3}.$$

相加得

$$\frac{\partial^2 r}{\partial x^2} + \frac{\partial^2 r}{\partial y^2} + \frac{\partial^2 r}{\partial z^2} = \frac{3r^2 - (x^2 + y^2 + z^2)}{r^3} = \frac{3r^2 - r^2}{r^3} = \frac{2r^2}{r^3} = \frac{2}{r}.$$

Q.E.D.

$$(2) \quad \frac{\partial^2 \ln r}{\partial x^2} + \frac{\partial^2 \ln r}{\partial y^2} + \frac{\partial^2 \ln r}{\partial z^2} = \frac{1}{r^2};$$

Proof. 一阶偏导数为

$$\frac{\partial \ln r}{\partial x} = \frac{1}{r} \cdot \frac{\partial r}{\partial x} = \frac{x}{r^2}.$$

因此二阶偏导数为

$$\frac{\partial^2 \ln r}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{x}{r^2} \right) = \frac{r^2 - x \cdot 2r \cdot \frac{\partial r}{\partial x}}{r^4} = \frac{r^2 - 2x^2}{r^4}.$$

同理可得

$$\frac{\partial^2 \ln r}{\partial y^2} = \frac{r^2 - 2y^2}{r^4}, \quad \frac{\partial^2 \ln r}{\partial z^2} = \frac{r^2 - 2z^2}{r^4}.$$

相加得

$$\frac{\partial^2 \ln r}{\partial x^2} + \frac{\partial^2 \ln r}{\partial y^2} + \frac{\partial^2 \ln r}{\partial z^2} = \frac{3r^2 - 2(x^2 + y^2 + z^2)}{r^4} = \frac{3r^2 - 2r^2}{r^4} = \frac{1}{r^2}.$$

Q.E.D.

$$(3) \quad \frac{\partial^2}{\partial x^2} \frac{1}{r} + \frac{\partial^2}{\partial y^2} \frac{1}{r} + \frac{\partial^2}{\partial z^2} \frac{1}{r} = 0.$$

Proof. 一阶偏导数为

$$\frac{\partial}{\partial x} \left(\frac{1}{r} \right) = -\frac{1}{r^2} \cdot \frac{\partial r}{\partial x} = -\frac{x}{r^3}.$$

二阶偏导数为

$$\frac{\partial^2}{\partial x^2} \left(\frac{1}{r} \right) = \frac{\partial}{\partial x} \left(-\frac{x}{r^3} \right) = -\frac{r^3 - x \cdot 3r^2 \cdot \frac{\partial r}{\partial x}}{r^6} = -\frac{r^3 - 3rx^2}{r^6} = \frac{3x^2 - r^2}{r^5}.$$

同理可得

$$\frac{\partial^2}{\partial y^2} \left(\frac{1}{r} \right) = \frac{3y^2 - r^2}{r^5}, \quad \frac{\partial^2}{\partial z^2} \left(\frac{1}{r} \right) = \frac{3z^2 - r^2}{r^5}.$$

相加得

$$\frac{\partial^2}{\partial x^2} \frac{1}{r} + \frac{\partial^2}{\partial y^2} \frac{1}{r} + \frac{\partial^2}{\partial z^2} \frac{1}{r} = \frac{3(x^2 + y^2 + z^2) - 3r^2}{r^5} = \frac{3r^2 - 3r^2}{r^5} = 0.$$

Q.E.D.

13. 求下列函数的微分, 或在给定点的微分

(2) $z = \frac{xy}{x^2 + y^2}$;

Solution. 一阶偏导数为

$$\frac{\partial z}{\partial x} = \frac{y(x^2 + y^2) - xy \cdot 2x}{(x^2 + y^2)^2} = \frac{y^3 - x^2y}{(x^2 + y^2)^2} = \frac{y(y^2 - x^2)}{(x^2 + y^2)^2}.$$

同理可得

$$\frac{\partial z}{\partial y} = \frac{x(x^2 - y^2)}{(x^2 + y^2)^2}.$$

因此

$$dz = \frac{\partial z}{\partial x}dx + \frac{\partial z}{\partial y}dy = \frac{y(y^2 - x^2)}{(x^2 + y^2)^2}dx + \frac{x(x^2 - y^2)}{(x^2 + y^2)^2}dy.$$

(4) $z = \arctan \frac{y}{x}$;

Solution. 一阶偏导数为

$$\frac{\partial z}{\partial x} = \frac{1}{1 + \left(\frac{y}{x}\right)^2} \cdot \left(-\frac{y}{x^2}\right) = -\frac{y}{x^2 + y^2},$$

$$\frac{\partial z}{\partial y} = \frac{1}{1 + \left(\frac{y}{x}\right)^2} \cdot \frac{1}{x} = \frac{x}{x^2 + y^2}.$$

因此

$$dz = -\frac{y}{x^2 + y^2}dx + \frac{x}{x^2 + y^2}dy.$$

(6) $z = x^4 + y^4 - 4x^2y^2$ 在点 $(0, 0), (1, 1)$;

Solution. 一阶偏导数为

$$\frac{\partial z}{\partial x} = 4x^3 - 8xy^2, \quad \frac{\partial z}{\partial y} = 4y^3 - 8x^2y.$$

在给定点处的值为

$$\begin{cases} \left. \frac{\partial z}{\partial x} \right|_{(0,0)} = 0 \\ \left. \frac{\partial z}{\partial y} \right|_{(0,0)} = 0 \end{cases}, \quad \begin{cases} \left. \frac{\partial z}{\partial x} \right|_{(1,1)} = 4 - 8 = -4 \\ \left. \frac{\partial z}{\partial y} \right|_{(1,1)} = 4 - 8 = -4 \end{cases}.$$

因此

$$\begin{cases} dz|_{(0,0)} = 0 \cdot dx + 0 \cdot dy = 0 \\ dz|_{(1,1)} = -4dx - 4dy \end{cases}.$$

15. 根据可微的定义证明, 函数 $f(x, y) = \sqrt{|xy|}$ 在原点处不可微。

Proof. 即证

$$\Delta z = f(\Delta x, \Delta y) - f(0, 0) = f_x(0, 0)\Delta x + f_y(0, 0)\Delta y + o(h),$$

其中 $h = \sqrt{(\Delta x)^2 + (\Delta y)^2} \rightarrow 0$ 。

计算偏导数：

$$f_x(0, 0) = \lim_{\Delta x \rightarrow 0} \frac{f(\Delta x, 0) - f(0, 0)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{0 - 0}{\Delta x} = 0,$$

$$f_y(0, 0) = \lim_{\Delta y \rightarrow 0} \frac{f(0, \Delta y) - f(0, 0)}{\Delta y} = 0.$$

因此

$$\lim_{h \rightarrow 0} \frac{\Delta z - [f_x(0, 0)\Delta x + f_y(0, 0)\Delta y]}{h} = \lim_{h \rightarrow 0} \frac{f(\Delta x, \Delta y) - f(0, 0)}{h} = \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \frac{\sqrt{|\Delta x \Delta y|}}{\sqrt{(\Delta x)^2 + (\Delta y)^2}}.$$

取 $\Delta y = \Delta x$ ，则

$$\lim_{\Delta x \rightarrow 0} \frac{\sqrt{|\Delta x \Delta y|}}{\sqrt{(\Delta x)^2 + (\Delta y)^2}} = \lim_{\Delta x \rightarrow 0} \frac{|\Delta x|}{\sqrt{2}|\Delta x|} = \frac{1}{\sqrt{2}} \neq 0.$$

因此函数在原点处不可微。

Q.E.D.

17. 证明函数

$$f(x, y) = \begin{cases} (x^2 + y^2) \sin \frac{1}{\sqrt{x^2 + y^2}}, & x^2 + y^2 \neq 0, \\ 0, & x^2 + y^2 = 0 \end{cases}$$

在点 $(0, 0)$ 连续且偏导数存在，但偏导数在点 $(0, 0)$ 不连续，而 f 在原点 $(0, 0)$ 可微。

Proof. 1. 连续性：由于

$$\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = \lim_{\sqrt{x^2 + y^2} \rightarrow 0} (x^2 + y^2) \sin \frac{1}{\sqrt{x^2 + y^2}} = f(0, 0),$$

因此函数在原点 $(0, 0)$ 连续。

2. 偏导数存在性：

$$f_x(0, 0) = \lim_{\Delta x \rightarrow 0} \frac{f(\Delta x, 0) - f(0, 0)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{(\Delta x)^2 \sin \frac{1}{|\Delta x|}}{\Delta x} = \lim_{\Delta x \rightarrow 0} \Delta x \sin \frac{1}{|\Delta x|} = 0.$$

同理可得 $f_y(0, 0) = 0$ ，因此偏导数在 $(0, 0)$ 处存在。

3. 偏导数不连续性：当 $\sqrt{x^2 + y^2} \neq 0$ 时，

$$f_x(x, y) = 2x \sin \frac{1}{\sqrt{x^2 + y^2}} - \frac{x}{\sqrt{x^2 + y^2}} \cos \frac{1}{\sqrt{x^2 + y^2}}.$$

令 $y = 0$, 则 $\sqrt{x^2 + y^2} = |x|$, 于是

$$f_x(x, 0) = 2x \sin \frac{1}{|x|} - \frac{x}{|x|} \cos \frac{1}{|x|}.$$

由于 $\lim_{x \rightarrow 0} \cos \frac{1}{|x|}$ 无限振荡, 极限不存在, 因此偏导数在点 $(0, 0)$ 不连续。

同理可证偏导数 $f_y(x, y)$ 在点 $(0, 0)$ 不连续。

4. 可微性: 即证

$$\Delta z = f_x(0, 0)\Delta x + f_y(0, 0)\Delta y + o(h),$$

其中 $h = \sqrt{(\Delta x)^2 + (\Delta y)^2}$ 。

由已证 $f_x(0, 0) = 0$, $f_y(0, 0) = 0$, 只需证明 $\Delta z = o(h)$ 。

由于

$$\Delta z = f(\Delta x, \Delta y) - f(0, 0) = h^2 \sin \frac{1}{h},$$

因此

$$\frac{\Delta z}{h} = h \sin \frac{1}{h} \rightarrow 0 \quad (h \rightarrow 0).$$

即 f 在原点 $(0, 0)$ 可微。

Q.E.D.

18. 对下列函数, 求出关于 x 和 y 的所有一阶和二阶 (含混合) 偏导数。

(1) $z = u \ln v$, 其中 $u = x^2$, $v = \frac{1}{1+y}$;

Solution. 化简得

$$z = x^2 \ln \frac{1}{1+y} = -x^2 \ln(1+y).$$

因此一阶偏导数为

$$\frac{\partial z}{\partial x} = -2x \ln(1+y), \quad \frac{\partial z}{\partial y} = -\frac{x^2}{1+y}.$$

从而二阶偏导数为

$$\frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} (-2x \ln(1+y)) = -2 \ln(1+y),$$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial y} (-2x \ln(1+y)) = -\frac{2x}{1+y},$$

$$\frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y} \left(-\frac{x^2}{1+y} \right) = \frac{x^2}{(1+y)^2}.$$

(2) $z = u \arctan v$, 其中 $u = \frac{xy}{x-y}$, $v = x^2y + y - x$.

Solution. 首先计算 z 关于中间变量的偏导数:

$$\frac{\partial z}{\partial u} = \arctan v, \quad \frac{\partial z}{\partial v} = \frac{u}{1+v^2}.$$

其次计算 u, v 关于 x, y 的一阶偏导数:

$$\frac{\partial u}{\partial x} = -\frac{y^2}{(x-y)^2}, \quad \frac{\partial u}{\partial y} = \frac{x^2}{(x-y)^2},$$

$$\frac{\partial v}{\partial x} = 2xy - 1, \quad \frac{\partial v}{\partial y} = x^2 + 1.$$

应用链式法则, 得一阶偏导数:

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial x} = -\frac{y^2}{(x-y)^2} \arctan v + \frac{(2xy-1)u}{1+v^2},$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial y} = \frac{x^2}{(x-y)^2} \arctan v + \frac{(x^2+1)u}{1+v^2}.$$

进而计算二阶偏导数:

$$\begin{aligned} \frac{\partial^2 z}{\partial x^2} &= \frac{\partial}{\partial x} \left(-\frac{y^2}{(x-y)^2} \arctan v \right) + \frac{\partial}{\partial x} \left(\frac{(2xy-1)u}{1+v^2} \right) \\ &= \frac{2y^2}{(x-y)^3} \arctan v - \frac{y^2}{(x-y)^2} \cdot \frac{1}{1+v^2} (2xy-1) \\ &\quad + \frac{(2y)(1+v^2)u + (2xy-1)(1+v^2) \frac{\partial u}{\partial x} - (2xy-1)u \cdot 2v \frac{\partial v}{\partial x}}{(1+v^2)^2} \\ &= \frac{2y^2}{(x-y)^3} \arctan(x^2y + y - x) - \frac{y^2}{(x-y)^2} \cdot \frac{1}{1+(x^2y + y - x)^2} (2xy-1) \\ &\quad + \frac{(2y)(1+(x^2y+y-x)^2)(\frac{xy}{x-y}) + (2xy-1)(1+(x^2y+y-x)^2) \frac{-y^2}{(x-y)^2} - (2xy-1)(\frac{xy}{x-y}) \cdot 2(x^2y+y-x)(2xy-1)}{(1+(x^2y + y - x)^2)^2}. \end{aligned}$$

$$\begin{aligned} \frac{\partial^2 z}{\partial y^2} &= \frac{\partial}{\partial y} \left(\frac{x^2}{(x-y)^2} \arctan v \right) + \frac{\partial}{\partial y} \left(\frac{(x^2+1)u}{1+v^2} \right) \\ &= -\frac{2x^2}{(x-y)^3} \arctan v + \frac{x^2}{(x-y)^2} \cdot \frac{1}{1+v^2} (x^2+1) \\ &\quad + \frac{(x^2+1)(1+v^2) \frac{\partial u}{\partial y} - (x^2+1)u \cdot 2v \frac{\partial v}{\partial y}}{(1+v^2)^2} \\ &= -\frac{2x^2}{(x-y)^3} \arctan(x^2y + y - x) + \frac{x^2}{(x-y)^2} \cdot \frac{1}{1+(x^2y + y - x)^2} (x^2+1) \\ &\quad + \frac{(x^2+1)(1+(x^2y+y-x)^2) \frac{x^2}{(x-y)^2} - (x^2+1)(\frac{xy}{x-y}) \cdot 2(x^2y+y-x)(x^2+1)}{(1+(x^2y + y - x)^2)^2}. \end{aligned}$$

$$\begin{aligned}
\frac{\partial^2 z}{\partial x \partial y} &= \frac{\partial}{\partial y} \left(-\frac{y^2}{(x-y)^2} \arctan v \right) + \frac{\partial}{\partial y} \left(\frac{(2xy-1)u}{1+v^2} \right) \\
&= -\frac{2y(x-y)^2 + 2y^2(x-y)}{(x-y)^4} \arctan v - \frac{y^2}{(x-y)^2} \cdot \frac{1}{1+v^2} (x^2+1) \\
&\quad + \frac{(2x)(1+v^2)u + (2xy-1)(1+v^2) \frac{\partial u}{\partial y} - (2xy-1)u \cdot 2v \frac{\partial v}{\partial y}}{(1+v^2)^2} \\
&= -\frac{2y(x-y)^2 + 2y^2(x-y)}{(x-y)^4} \arctan(x^2y+y-x) - \frac{y^2}{(x-y)^2} \cdot \frac{1}{1+(x^2y+y-x)^2} (x^2+1) \\
&\quad + \frac{(2x)(1+(x^2y+y-x)^2)(\frac{xy}{x-y}) + (2xy-1)(1+(x^2y+y-x)^2) \frac{x^2}{(x-y)^2} - (2xy-1)(\frac{xy}{x-y}) \cdot 2(x^2y+y-x)(x^2+1)}{(1+(x^2y+y-x)^2)^2}.
\end{aligned}$$